

# SciOdyssey.

Autonomous ML Research Agent

Preserve the research world.  
Reset the researcher.

LONGER CONTEXTS. BRIGHTER SCIENCE.  
SAME WORLD. MORE MINDS.

## 01 / PROBLEM

Different lifetimes.  
Shared evidence.

Long-lived reasoning can anchor search. Tree search rereads files that cost more tokens to understand than edit. SciOdyssey retains the world and restarts with a fresh Scientist.



KNOWLEDGE  
OUTLIVES  
INDIVIDUALS.

## 02 / MOTIVATION

### Research world

PERSISTENT  
ON DISK

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| EVIDENCE | Knowledge + experiment ledger<br>Metrics, failures and original reports         |
| CURATED  | Research brief + fallible priors<br>Current understanding, open to revision     |
| CODE     | Retained implementation + State<br>Recoverable version, evidence and provenance |

Context reset — inherit experience, not prior reasoning.

## 03 / SYSTEM

### Scientist

SCIENTIFIC JUDGMENT

- Read the research world
- Form hypotheses
- Code, train and evaluate
- Interpret and report evidence

Chooses the science within one trajectory.

### META

PERSISTENCE AUTHORITY

- Audit claims against evidence
- Scope conclusions
- Curate shared memory
- Preserve optional State

Maintains what survives across trajectories.

Audit → scope → preserve → fresh Scientist

## 04 / EXPERIENCE

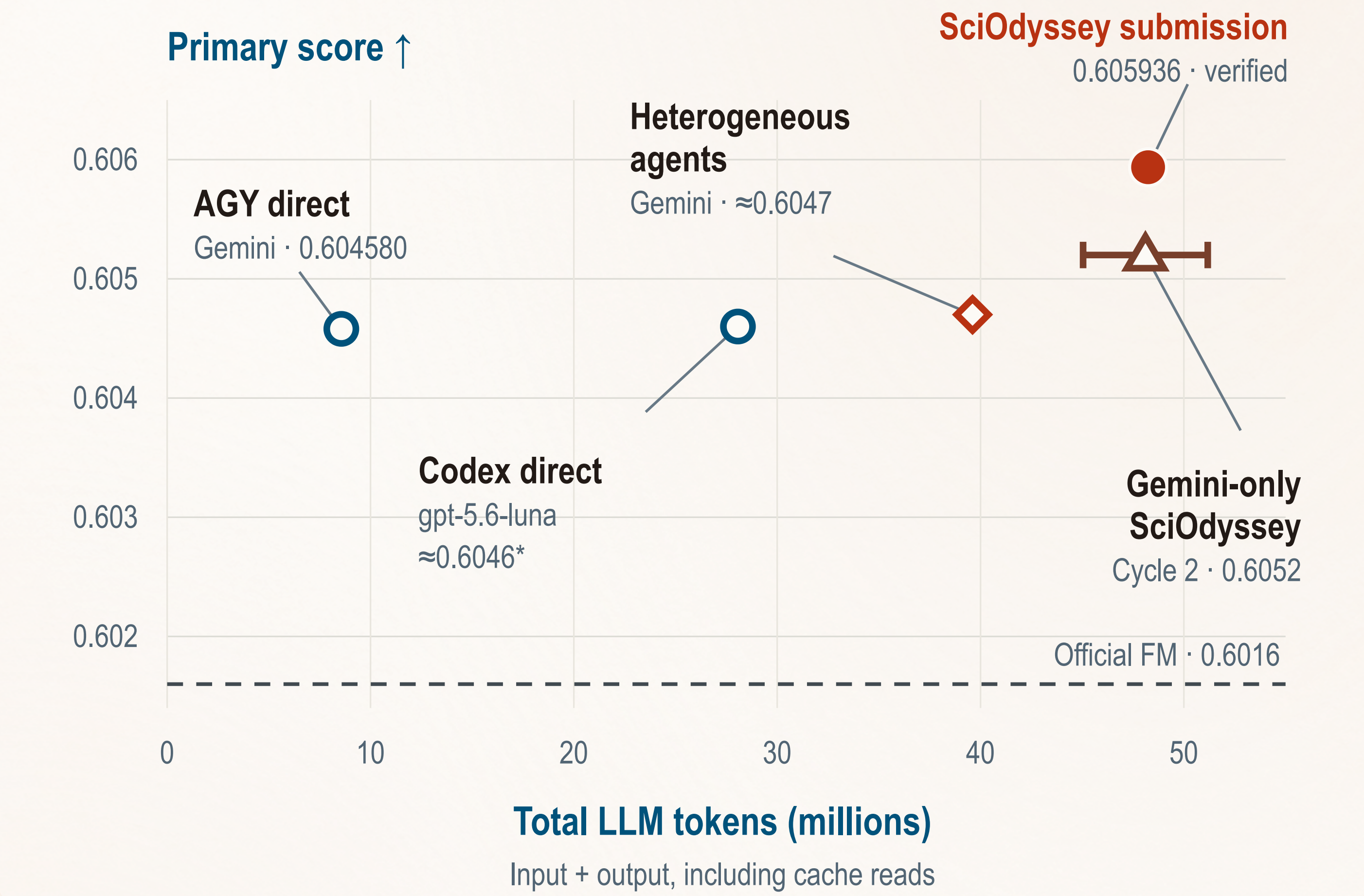
One agent. Any ML task. End to end.

|                                                                                                                               |                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| task.md<br>What to solve<br>Objectives · constraints · evaluation                                                             | PERSONAL.md<br>How you want it solved<br>Preferences · research style · priorities |
| Define the task →                                                                                                             | Delegate research → Review evidence                                                |
| EXAMPLE INPUT                                                                                                                 |                                                                                    |
| Use the research-agent skill for task.md.<br>Follow PERSONAL.md. Run autonomously.<br>Let me review the results and evidence. |                                                                                    |

## 05 / EVALUATION

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



\* Codex is provisional. Gemini = gemini-3.7-flash. The Cycle-2 interval is a token-accounting bound. Models and protocols differ: no causal, significance or token-efficiency claim is made.

### Verified submission · public validation

| Metric  | Official FM | SciOdyssey | Δ          |
|---------|-------------|------------|------------|
| Primary | 0.6016000   | 0.6059363  | +0.0043363 |
| GAUC    | 0.6674000   | 0.6728421  | +0.0054421 |
| nDCG@5  | 0.5357000   | 0.5390304  | +0.0033304 |

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features, eight-seed FM ensemble, NumPy / CPU.

4 autonomous cycles | 13 named experiments | 0 manual scientific interventions\*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

\* After launch, in the retained submission record.

## 06 / INSIGHTS

Observed behavior, not just a score.

|                                                                                                                                            |                                                                                                                   |                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recovery</b><br>NumPy float32 values broke serialization after evaluation. The Scientist repaired the writer and retained the evidence. | <b>Audit</b><br>Reviews found leakage and sampling mismatches. META scopes evidence; it is not a formal verifier. | <b>Open questions</b><br>One task does not establish generality. META, State and reset effects are not isolated. Summaries can lose context and anchor later work. |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

BETTER PROCESSES.  
A WIDER RESEARCH WORLD.